DESIGN REPORT

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GE1501 | 1:35 PM

Abstract (178)

Tortoises face life-threatening risks when trapped on their backs including lung pressure that restricts breathing and, in hot climates, dangerous temperature spikes leading to dehydration and heat stress. Tortoises with conditions like buoyancy disorders, shell fractures, or float syndrome are at an even higher risk of being unable to self-right. These challenges are compounded for tortoises with flat or saddleback-shaped shells, which make it harder to regain balance on their own.

D.O.U.G - otherwise known as the Dynamic Overturning and Uprighting Gizmo - aims to help tortoises, specifically those in Cerrado savannas of Brazil or the West African savannas, get back onto their feet. The device would be designed to attach securely to the tortoise's shell and use a pressure sensor to detect when the tortoise is inverted, ensuring reliable activation without relying on traditional buttons. A 180-servo motor would deploy a small, weighted lever to gently flip the tortoise back onto its feet. The design would account for the tortoise's natural behaviors, such as digging tunnels, ensuring the device is durable and resilient under rough conditions.

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Brief Reflection (241)

Our team employed the five steps of the engineering design process to develop the Dynamic Overturning and Uprighting Gizmo (D. O. U. G.) for tortoises. First we identified the problem by recognizing that most tortoises, when stuck on their shells, are unable to right themselves. When overturned, their organs press onto their lungs making it difficult to breathe in that orientation. While the geometry of their shells allows them some momentum to overturn, it is not enough so we started brainstorming. Through our decision matrix we explored five different design concepts: flip device, spring, shell dome, an airbag, and a stick on the shell. We had multiple criteria but prioritized even weight distribution of the device as well as durability. Once we decided on the flip device, there were a couple iterations and ideas that we discarded due to lack of material and time before settling on the current design. Once we settled on our design, we began prototyping and testing. We used an Arduino Uno and breadboard to test and perfect our circuit before soldering it onto a perf board to save space and weight. There was a lot of testing required with the circuit post solder to ensure that it wasn't shorting and that the servo was still receiving power. That was where we struggled the most. In the future we plan to create a more reliable and compact circuit by using an Arduino Nano and better soldering skills.

Decision Matrix/ Decision Making Tools:

Decision Matrix						
Objectives	1: Shell Size	2: Distributed Weight	3: Durable	4: Aesthetics	5: Comfort	
Weights:	50	90	80	20	60	
	Rating (scale of 1-10) / Rating x Weight					TOTAL
Design 1: Flip Device	350	810	720	0	350	2230
Design 2: Spring	100	N/A	800	0	480	1380
Design 3: Shell Dome	500	630	480	0	360	1970
Design 4: Air Bag	250	180	560	0	540	1530
Design 5: Stick on Shell	100	180	640	0	540	1460

Design #1 Flip Device won with 2230 points, with Design #3 Shell Dome and Design #4 Air Bag coming in second and third place respectively. We picked five critical objectives such as shell size, weight distribution, durability, aesthetics, and comfort. We also discussed the feasibility of each and determined the flip device to be most aligned with the project requirements. (60)

Expenses Breakdown

Super Glue	$2.00
9V Battery	$5.00
Pressure Sensor	$7.45
Tri Fold Poster Board	$11.99
	$26.44

Parts List

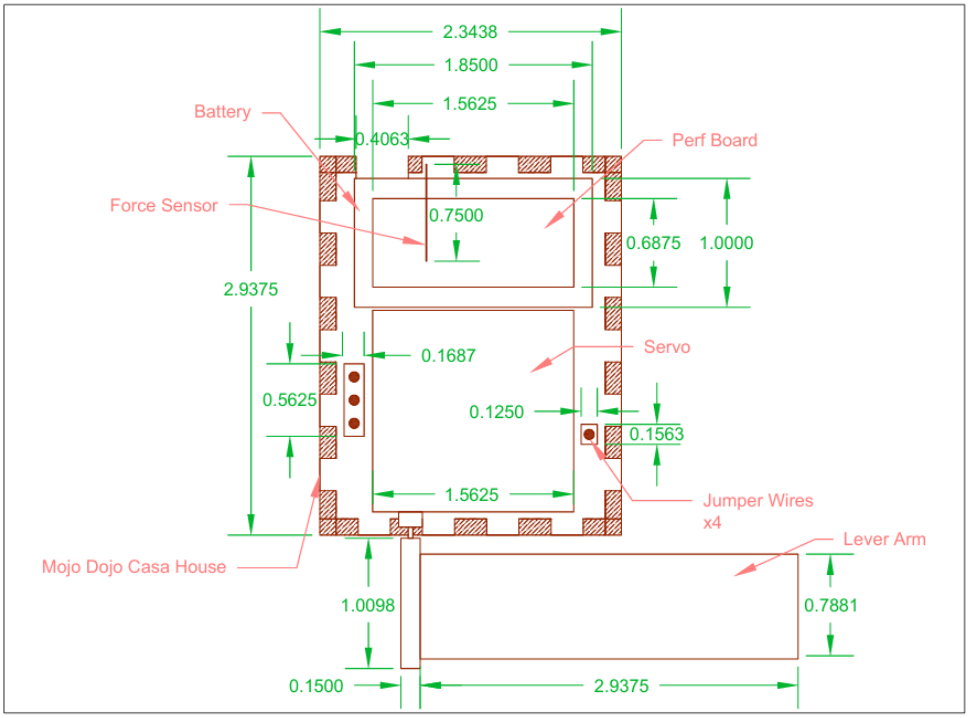
Electronics	Materials
• 9V Battery • Arduino Uno Redboard • Jumper Wires • Force Sensor • Perf Board • MG995 Servo Motor • 330 Ohm Resistor	• 1/8th inch MDF Wood • 1/8th inch Plywood • 3D Print PLA Filament • Fabric • Velcro

Gantt Chart

	Week of 10/27	Week of 11/03	Week of 11/10	Week of 11/17	Week of 11/24	Week of 12/01
Prototype #1						
Prototype #2						
Final						
Design Report						

Orthographic Drawings

TOP



Top View

December 5, 2024

Scale

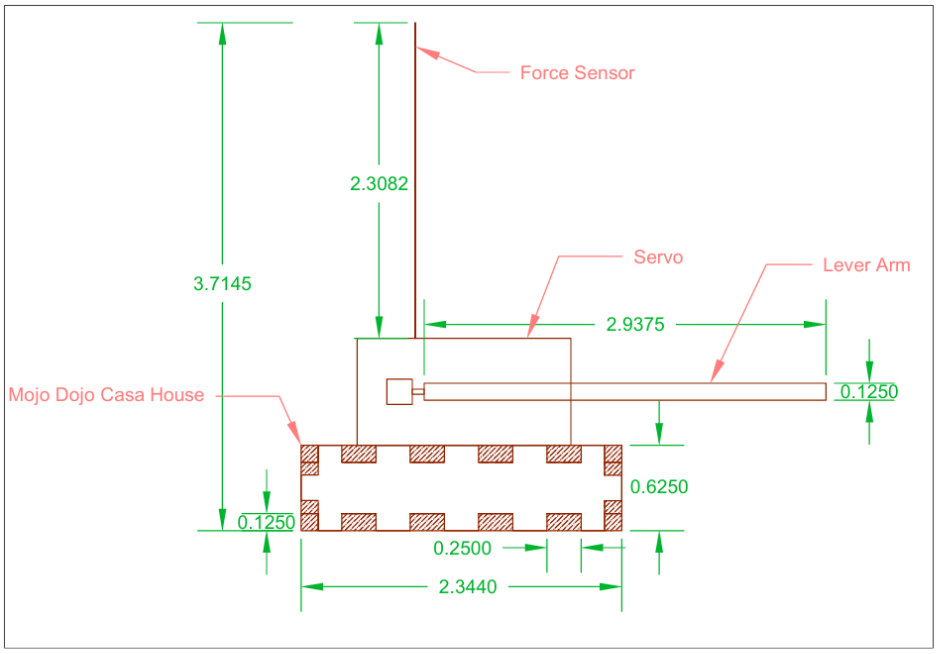
1:1

DOUG

(Dynamic Overturning and Uprighting Gizmo)

Aarushi, Jasmine, David, Andrea

FRONT



Front View

December 5, 2024

Scale

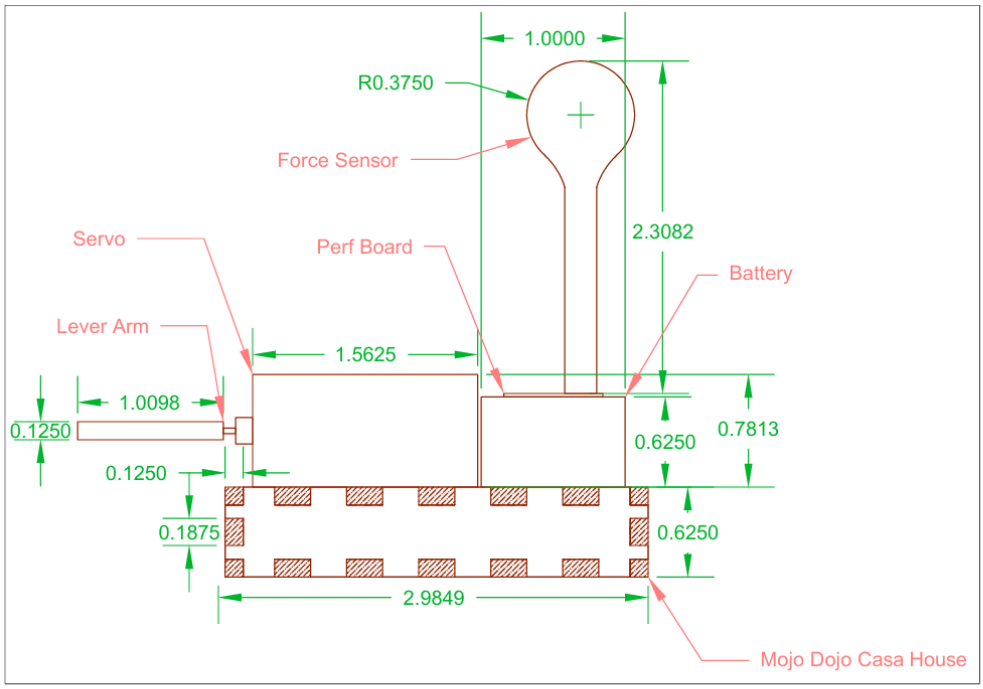
1:1

DOUG

(Dynamic Overturning and Uprighting Gizmo)

Aarushi, Jasmine, David, Andrea

SIDE



Side View

December 5, 2024

Scale

1:1

DOUG

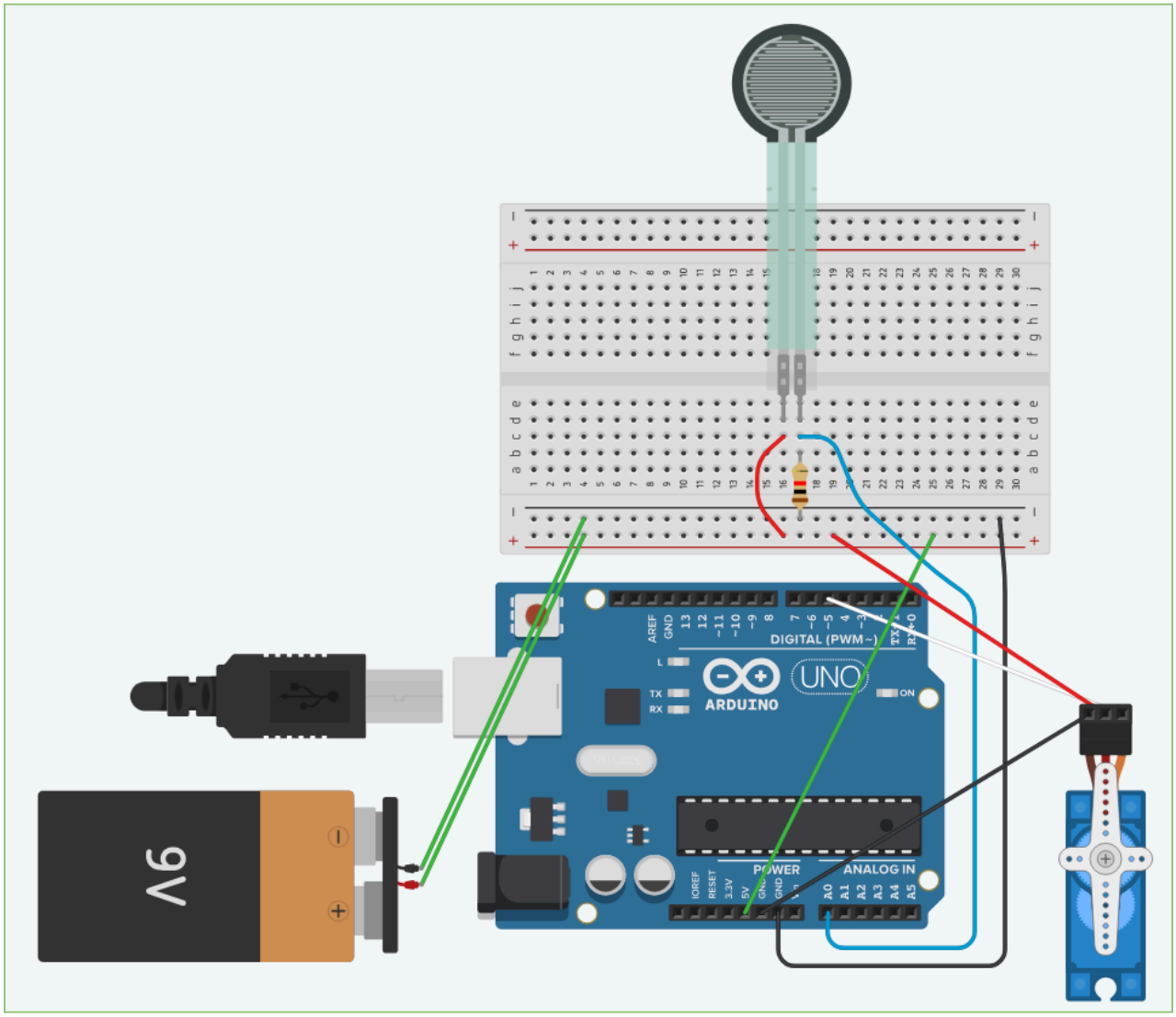
(Dynamic Overturning and Uprighting Gizmo)

Aarushi, Jasmine, David, Andrea

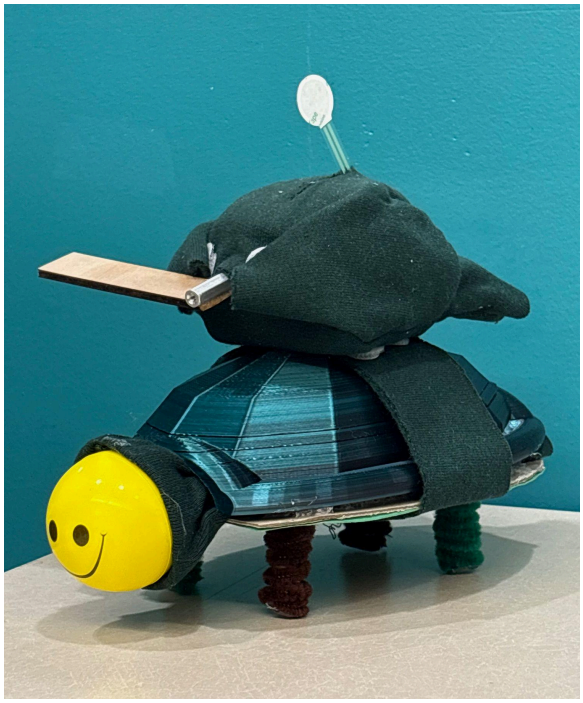
Initial Sketch

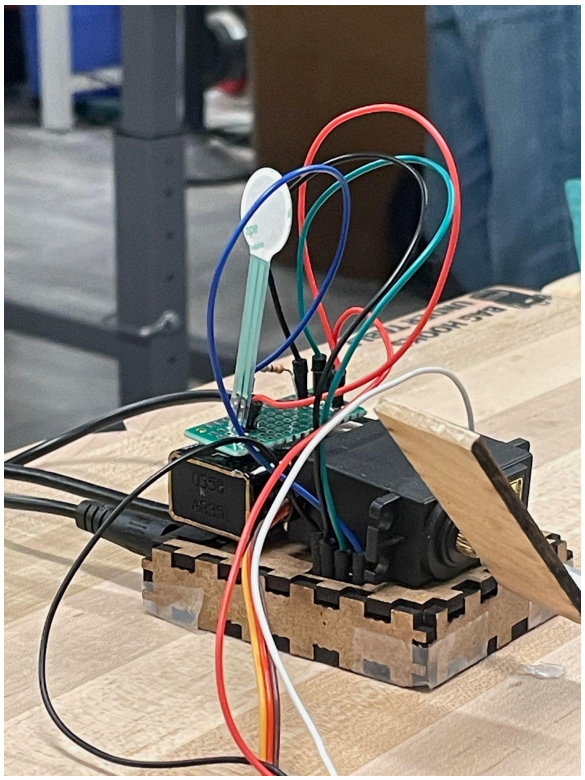
The original design was book shaped with the entire lid mechanized to lift up performing the same action that our current lever arm does. We then figured we could simplify our design to just the lever arm and discarded this idea. We also had planned on using TPU filament for the base due to its flexible nature instead of a backpack design. However, due to a lack of TPU and a printer that could print TPU, we redesigned the circuit to fit in a backpack. (85)

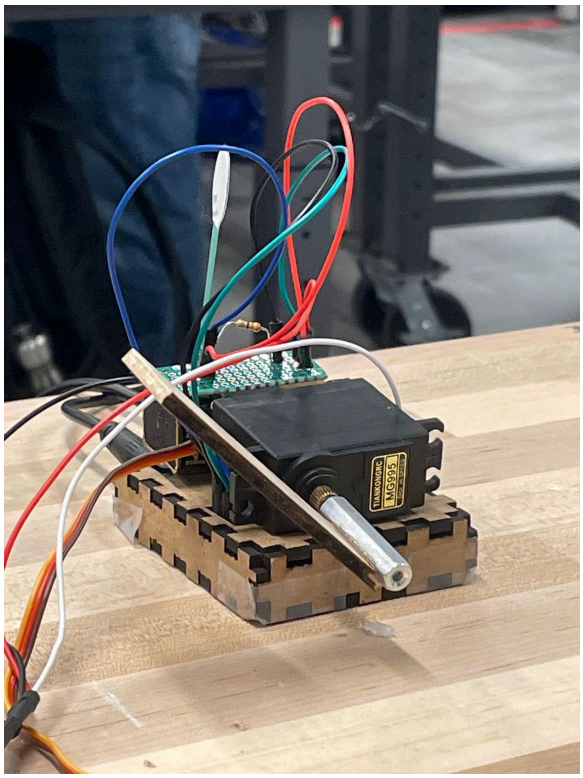
Wiring Diagram (TinkerCad)



Photographs







Future Work Statement (152)

In future iterations, we hope to solve any flaws in our current design while also introducing changes to improve functionality and usability. Cleaner and more exact soldering might increase the circuit's reliability. To achieve a more compact and lightweight design, we intend to use an Arduino Nano and optimize the device's form factor, ensuring that it is no heavier than the tortoise's shell while also offering adequate protection. In addition, we will aim to make the device sleeker and more user-friendly by improving access to critical components, such as the Arduino's reset button, which is currently unreachable due to design constraints caused by the backpack fabric. We also intend to include a Bluetooth module for wireless control, utilize a more powerful servo motor for larger tortoises, improve weather resistance with stronger, water-resistant materials, and increase comfort with filament or foam cushioning. These enhancements will make the gadget more efficient, durable, and adaptable to different tortoise sizes and conditions.

An Appendix

```cpp
#include <iostream>
using namespace std;
int main() {
    int PlayAgain = true; //Set this value initially to true so the loop runs
    int guide;
    while (PlayAgain) {      //things in the loop
//initial options for the user to look further into
        cout << "What would you like to know about the Dynamic Overturning Uprighting Gizmo (DOUG)? " << endl;
        cout << "Type '1' for a general description" << endl;
        cout << "Type '2' for instructions on how to use the device" << endl;
        cout << "Type '3' for material information" << endl;
        cout << "Type '4' for frequently asked questions and answers" << endl;
        cin >> guide;
        switch (guide) {
            case 1:     //general description of device
                cout << "The Dynamic Overturning Up-righting Gizmo works through a force sensor " << endl;
            cout << "attached to the shell of a small to medium-sized tortoise. " << endl;
            cout << "When the tortoise ends up upside down on its shell, " << endl;
            cout << "the force sensor will be triggered, and the servo inside " << endl;
            cout << "the device will automatically lift the tortoise back on its feet." << endl;
            break;

            case 2:     //general instructions for the device
                cout << "The device comes with a strap to attach to the bottom of the " << endl;
            cout << "tortoise to secure it. Close the velcro on the bottom and top of the " << endl;
            cout << "backpack so nothing can fall off. Make sure there are batteries attached " << endl;
            cout << "to the device; it should now work. When the tortoise is flipped over, " << endl;
            cout << "the device will automatically wait five seconds to let the tortoise start " << endl;
            cout << "the flipping process before triggering the servo and keeping it " << endl;
            cout << "in a 180-degree position for 20 seconds. After 20 seconds " << endl;
```

```cpp
            cout << "the servo will reset to 0 degrees. If the device malfunctions hit " << endl;
            cout << "the reset button or reenter code into the circuit." << endl;
            break;
            case 3:     //material information for device
                cout << "PARTS:" << endl;
                cout << "9V Battery" << endl;
                cout << "Arduino Uno Board" << endl;
                cout << "Jumper Wires" << endl;
                cout << "Force Sensor" << endl;
                cout << "Perf Board" << endl;
                cout << "MG995 Servo Motor" << endl;
                cout << "330 Ohm Resistor" << endl;
                cout << "MATERIALS:" << endl;
                cout << "1/8th inch MDF Wood" << endl;
                cout << "1/8th inch Plywood" << endl;
                cout << "Velcro" << endl;
                cout << "3D Printed Shell" << endl;
                cout << "Fabric" << endl;
            break;
            case 4:     //QNA for device
                cout << "Question 1: What can happen to my tortoise if it's on its back for too long?" << endl;
            cout << "Answer: A tortoise being on its back for too long can be deadly due to overheating from the sun, "
                    "lung damage from organ compression, or starvation. " << endl;
            cout << "Question 2: Can DOUG work for any size tortoise?" << endl;
```

```
            cout << "Answer: Currently DOUG can only lift tortoises up to 2kg but we are working on incorporating more "
                    "powerful servo motors into the device to accommodate larger tortoises." << endl;
            cout << "Question 3: Will the device be comfortable for a tortoise?" << endl;
            cout << "Answer: We know that tortoises can feel their shells and accordingly worked to make DOUG as comfortable "
                    "for the tortoise as possible. In the future we would like to increase the comfort even more by "
                    "implementing foam or filament into the design." << endl;
            break;
            default:        //default answer
                cout << "Invalid input" << endl;
        }
        char answer;
        const char yes = 'Y', no = 'N'; //declare constant variables
        //ask if user wants to go again
        cout << "\n"
            "Do you want to go back? Enter 'Y' for yes or 'N' for no:" << endl;   //ask user to play again
        cin >> answer;
        if (toupper(answer) == yes) {                  //Restarts the program
            PlayAgain = true;
        }
        else if (toupper(answer) == no) {              //Ends the program and prints message
            PlayAgain = false;
            cout << "Program ended.";
        }
```

```
        else {          //default answer
            PlayAgain = false;
            cout << "Sorry, " << answer << " is an invalid input\n";
        }
    }
}
```

The Code

```
#include <Servo.h> //needed for servo commands

int sensor = 0;   //Set servo to default position

Servo servo_5;    //declare servo

void setup()
{
  pinMode(A0, INPUT);   //set A0 as the input for the force sensor
  servo_5.attach(5, 500, 2500);
  Serial.begin(9600);
}

void loop()
{
  sensor = analogRead(A0);  //sensor reads the input
if (sensor > 20) {        //set to a low force value so it is picked up every time
    delay(5000); // Wait for 5000 millisecond(s)

    // Move servo from 15 to 150 in steps of 5
    for (int angle = 15; angle <= 150; angle += 5) {
        servo_5.write(angle);
        delay(40);
    }

    delay(20000); // Wait for 20000 millisecond(s)

    // Move servo back from 145 to 0 in steps of 5
    for (int angle = 150; angle >= 0; angle -= 5) {
        servo_5.write(angle);
        delay(40);
    }
  } else {
    servo_5.write(0);   //keeps the servo in default position
  }
  Serial.print("sensor = ");      //shows the sensor value on serial monitor
  Serial.println(sensor);
  delay(100); // Wait for 100 millisecond(s)
}
```

Bibliography

N. Frost, "How a Flipped-Over Giant Tortoise Gets Its Feet Back on the Ground," Atlas Obscura, Dec. 2017. https://www.atlasobscura.com/articles/giant-tortoise-galapagos-flipped-turn-over (accessed Oct. 29, 2024).
Summary: This article discusses how different shelled Galápagos giant tortoises right themselves after flipping over. Tortoises with dome shaped shells have an easier job flipping over by "flinging their limbs back and forth" while saddleback tortoises require more energy to complete the process.
Relevance: We used this article to determine the exact species of tortoise we would target the product for since saddleback tortoises have a harder job generally. It also helped us learn the exact behavior of different tortoises when attempting to flip over.

Kamp Kenan, "Will a FLIPPED Tortoise SUFFOCATE?," YouTube, Nov. 17, 2018. https://www.youtube.com/watch?v=XK4v25sZAxA (accessed Oct. 29, 2024).
Summary: This video explains the dangers of tortoises being flipped over for extended periods, including breathing restrictions and overheating risks.
Relevance: We used this to quantify how long a tortoise can be flipped over, how big of an issue tortoises being flipped over is, and for general background information on the problem we were solving.

"What is the operating temperature range for Arduino boards?," Arduino Help Center. https://support.arduino.cc/hc/en-us/articles/360016076980-What-is-the-operating-temperature-range-for-Arduino-boards (accessed Nov. 2, 2024).
Summary: The article details the temperature limits within which Arduino boards can safely operate, typically ranging from -40°C to 85°C.
Relevance: This information was useful because we needed DOUG to work in the sun since that is when tortoises are most at risk. From the article we concluded that the arduino boards would be fine in almost all realistic temperatures.

"Thermal Properties - Performance Panels," Performancepanels.com, 2018. https://www.performancepanels.com/thermal-properties (accessed Nov. 2, 2024).
Summary: This resource outlines the thermal resistance and durability of various panels, including plywood and MDF.
Relevance: Helped in selecting appropriate materials for constructing the device, ensuring it can withstand extreme temperatures and environmental stresses.

G. Iacono, “Can Tortoises & Turtles Feel Their Shell? Vet-Verified Science & Info | PangoVet,” PangoVet, Sep. 18, 2024.
https://pangovet.com/pet-lifestyle/turtles/can-tortoises-turtles-feel-their-shell/ (accessed Nov. 2, 2024).
Summary: The article confirms that tortoises can feel touch on their shells due to nerve endings embedded in the bone.
Relevance: This information changed our outlook on the need for comfort since we didn’t previously know they had feelings in their shells. It led to discussions on the implementation of soft foam or filament in the design.

San Diego Zoo, “Turtles and tortoises,” San Diego Zoo Wildlife Explorers.
https://sdzwildlifeexplorers.org/animals/turtles-and-tortoises (accessed Nov. 2, 2024).
Summary: This page provides general information about the anatomy, behaviors, and habitats of turtles and tortoises.
Relevance: Offered foundational knowledge on tortoises that guided the development of a device compatible with their physiology and lifestyle.

Worldgineer, “Halfbakery: Tortoise Flipper,” Halfbakery.com, Mar. 25, 2003.
https://www.halfbakery.com/idea/Tortoise_20Flipper (accessed Oct. 29, 2024).
Summary: This conceptual idea suggests a mechanical flipper for tortoises but lacks detailed designs or implementation.
Relevance: This gave us inspiration when thinking of how to solve our problem statement and a foundation to build off of.

“MN1604 Size: 9V (6LR61) Alkaline-Manganese Dioxide Battery.” Available:
https://batteryspecialties.com/assets/pdf/duracell-mn1604-specification-sheet.pdf (accessed Nov. 14, 2024).
Summary: This specification sheet provides details on the capacity, voltage, and operating conditions of 9V batteries.
Relevance: We wanted to learn about the battery, make sure it was okay to use, and compare its weight to the previous battery pack we were using to see if there was a notable difference.

Adafruit Industries, “Tiny Premium Breadboard,” Adafruit.com, 2022.
https://www.adafruit.com/product/65#description (accessed Nov. 14, 2024).
Summary: Describes the features and specifications of a compact breadboard ideal for prototyping small-scale electronics.
Relevance: We considered using a mini breadboard for our device given that our current one was too big and wanted to know more about its specifications. We didn’t end up using it as we soldered, but it was an idea we considered during prototype 2.

G. Domokos and P. L. Várkonyi, “Geometry and self-righting of turtles,” The Royal Society, vol. 275, no. 1630, Nov. 2007.
https://europepmc.org/backend/ptpmcrender.fcgi?accid=PMC2562404&blobtype=pdf (accessed Oct. 29, 2024).
Summary: This paper talks about turtle shell geometry and how the natural shape of the shell helps them to self-right.
Relevance: This paper helped us better understand how these animals survive in nature. We used this information to act as an aid to their intrinsic movements instead of overriding instincts and completely mechanizing their shell.